

Started on	Wednesday, 27 November 2024, 2:17 PM
State	Finished
Completed on	Wednesday, 27 November 2024, 2:34 PM
Time taken	17 mins 4 secs
Marks	3.00/3.00
Grade	3.00 out of 3.00 (100%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of t tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you u the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final**

The folloing rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

If p and q be prime numbers, let $n = p \cdot q$, and write (n, x) and (n, y) for the public and private key in an RSA key associated to n .

In the list below we have written suggestions for $(x, y, p, q, p \cdot q)$. Some of them give valid RSA keys, some of them different reasons.

Select all options that give us a valid key pair. Leave the invalid options un-checked.

- ☐ $x = 23095092, y=2157127, p=7027, q=3917, n=27524759$
- ☒ $x = 20271383, y=22663367, p=3557, q=7741, n=27534737$
- ☐ $x = 35131943, y=15273131, p=7579, q=7583, n=57471557$
- ☐ $x = 7637383, y=15705847, p=3921, q=5531, n=21687051$
- ☒ $x = 3975723, y=862787, p=7901, q=3209, n=25354309$
- ☐ $x = 2124966, y=1889417, p=5477, q=2437, n=13347449$
- ☐ None of the above

You are free to use this calculator for products $\text{mod } (n)$. For example, if you want to compute $(7634 \cdot 99833 \% 8883993)$ should set **Factor 1**=7634, **Factor 2**=99382, and **Modulus**=8883993.

Factor 1:	
factor 2:	
Modulus:	
Result:	

You are free to use this calculator for computing powers $\text{mod } (n)$. For example, if you want to compute $(7634^{99382} \% 8883993)$, you should set **Base**=7634, **Exponent**=99382, and **Modulus**=8883993.

Base:	
Exponent:	
Modulus:	
Result:	

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

[000110, 000101, 000000, 101010]

.

$l_{min} =$

Your last answer was interpreted as follows:

2

$l_{max} =$

Your last answer was interpreted as follows:

5

Question **3**

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be the encoding function given by $E(w_1 w_2 w_3) = w_1 w_2 w_3 w_4 w_5$, where

$$\begin{bmatrix} w_4 = 0 \\ w_5 = w_2 \end{bmatrix}$$

a) Find the generator matrix G for the encoding E

$G =$

1 0 0 0 0
0 1 0 0 1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

b) Find the parity check matrix H for the encoding E .

$H =$

0 0 0 1 0
0 1 0 0 1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

c) Let $r = 01001$ be a received message. Compute the syndrome of r with respect to the encoding E .

Syndrome of $r =$

Your last answer was interpreted as follows:

00

Syntax guide: Give your answers to **a)** and **b)** as matrices. Use a space character to separate entries in the same row and a new line to break to separate rows.

Syntax guide: Give your answer to **c)** as a bitstring. E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of th

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 456

Signing code:

Your last answer was interpreted as follows:

12739

[◀ Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers m
Lagrange's theorem\)](#)

Jump to...